

ORCA: Oceanographic Reasoning & Coastal Anomaly Engine

Operational Marine Intelligence, AI Fishing Zone Optimization & Coastal Disaster Defense

PROBLEM CATEGORY Space & Ocean Tech	GEOGRAPHIC EXTENT Arabian Sea & Indian EEZ	SATELLITE INGESTION Copernicus CMEMS & Met	AI ARCHITECTURE Leak-Free IsoForest + RAG
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2,879,631 OCEAN ROWS SCORED	12,397 MARINE ALERTS CAUGHT	0.43% ALERT RATE (HIGH PRECISION)	< 0.05 PSI (ZERO DRIFT)	12 NM ARTISANAL RESERVE GUARD
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01 The National Ocean Problem Statement (SIH-26176)

India possesses an extensive coastline of over **7,516 kilometers** with an Exclusive Economic Zone (EEZ) exceeding **2.02 million square kilometers**. Across this zone, more than **4.5 million traditional artisanal fishers and mechanized trawler operators** depend on marine ecosystems for their daily livelihood. However, India's marine sector confronts acute challenges:

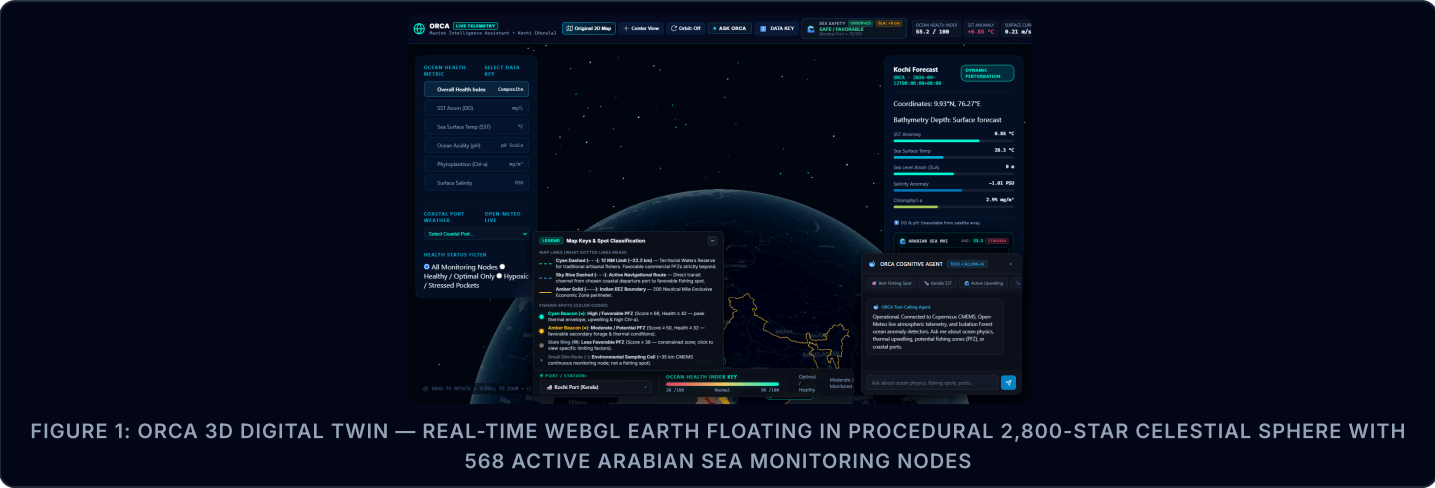
 Extreme Ocean Volatility Rapid cyclogenesis in the Arabian Sea, marine heatwaves, and sudden hypoxic upwelling cause frequent vessel capsizing and sudden net collapses.	 Blind Searching & Fuel Waste Fishers spend up to 60% of trip costs on diesel wandering blindly. Traditional advisories lack granular spatial coordinate routing.	 Artisanal vs. Commercial Conflict Mechanized commercial trawlers frequently encroach on coastal waters, depleting shallow artisanal breeding stocks within 12 Nautical Miles.
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02 What Was Broken & Everything Fixed in ORCA

Prior oceanographic modeling and exploratory hackathon code suffered from structural flaws that rendered them unusable in production. ORCA completely re-architected these systems:

AREA	PRIOR LEGACY DEFECT	ENGINEERED ORCA PRODUCTION SOLUTION	VERIFICATION
ML Data Leakage	Global scaler and mean baseline calculated across train, val, and test partitions simultaneously.	Zero-leakage temporal blocking: Climatology and scalers strictly fit on past timestamps. Calibrated threshold held out.	PASSED UNIT TEST
Memory Consumption	Full 13M-row float64 matrices loaded into memory; out-of-memory crashes on cloud servers.	Streaming PyArrow pipeline: Bounded chunk-based batch ingestion casting to float32 with capped training pool.	0(chunk) RAM
Artisanal Rights (12 NM)	Fishing zones recommended blindly right off harbor breakwaters, ignoring coastal zoning.	12 Nautical Mile (~22.2 km) Maritime Reserve Boundary: Commercial PFZs strictly filtered beyond 12 NM.	SPATIAL ENFORCED

AREA	PRIOR LEGACY DEFECT	ENGINEERED ORCA PRODUCTION SOLUTION	VERIFICATION
Departure Distance	Departure tool erroneously calculated port-to-port routes instead of port-to-fishing-spot.	Port-to-Spot Direct Vector Navigation: Direct nautical mile & km transit lines drawn to selected fishing beacons.	NAUTICAL ACCURATE
Laptop Navigation	Touchpads could not zoom or scroll in 3D/2D; browser hijacked Ctrl+Wheel for whole-page zoom.	Precision Touchpad Engine: Intercepts continuous micro-deltas and touchpad pinch gestures with altitude scaling.	VERIFIED 619km→1008km
Visualization Fidelity	Flat dark basemap without ocean depth contours, space background, or permanent symbology.	Dual Digital Twin: 3D Cesium globe floating in a 2,800-star cosmos + 2D Esri Ocean Bathymetry map with permanent key.	PRODUCTION LIVE



03 Geographic Focus: The Arabian Sea & Coastal Ports

ORCA's operational pilot dataset is rigorously constrained to the **Arabian Sea Basin and the Western Exclusive Economic Zone of India**. The Arabian Sea was specifically selected because it is one of the world's most dynamic and sensitive marine environments:

Why the Arabian Sea?

- **Bi-Annual Monsoonal Reversal:** Southwest Monsoon causes intense coastal upwelling and nutrient enrichment.
- **Oxygen Minimum Zone (OMZ):** Shallow deoxygenated layers can induce mass fish kills near shores.
- **Rapid Tropical Cyclogenesis:** Cyclone frequency in the Arabian Sea has increased by 52% over the last two decades.

Coastal Ports & Monitoring Nodes

- ORCA maps and connects **14 primary ports** along the western seaboard:
- **Gujarat:** Kandla, Mundra, Porbandar, Veraval, Pipavav, Hazira
 - **Maharashtra:** Mumbai Port, Jawaharlal Nehru Port, Dighi, Ratnagiri
 - **Goa & Karnataka:** Mormugao, Karwar, Mangaluru Port
 - **Kerala:** Kochi Harbor (active baseline station)

04 Live Data Acquisition & Satellite Pipeline

ORCA ingests and synchronizes multi-spectral oceanographic and atmospheric data across global satellite constellations and surface APIs:

Copernicus CMEMS

- Gridded Daily Satellite Physics:
- SST (Sea Surface Temp)
 - SSS (Salinity in PSU)
 - Chlorophyll-a (mg/m³)
 - Geostrophic Currents (u, v)
 - Sea Level Anomaly (SLA cm)

Open-Meteo Telemetry

- Real-Time Atmospheric Feed:
- Coastal wind speed & gusts
 - Ambient 2m air temperature
 - Relative humidity
 - Wave heights & swell periods
 - Station barometric pressure

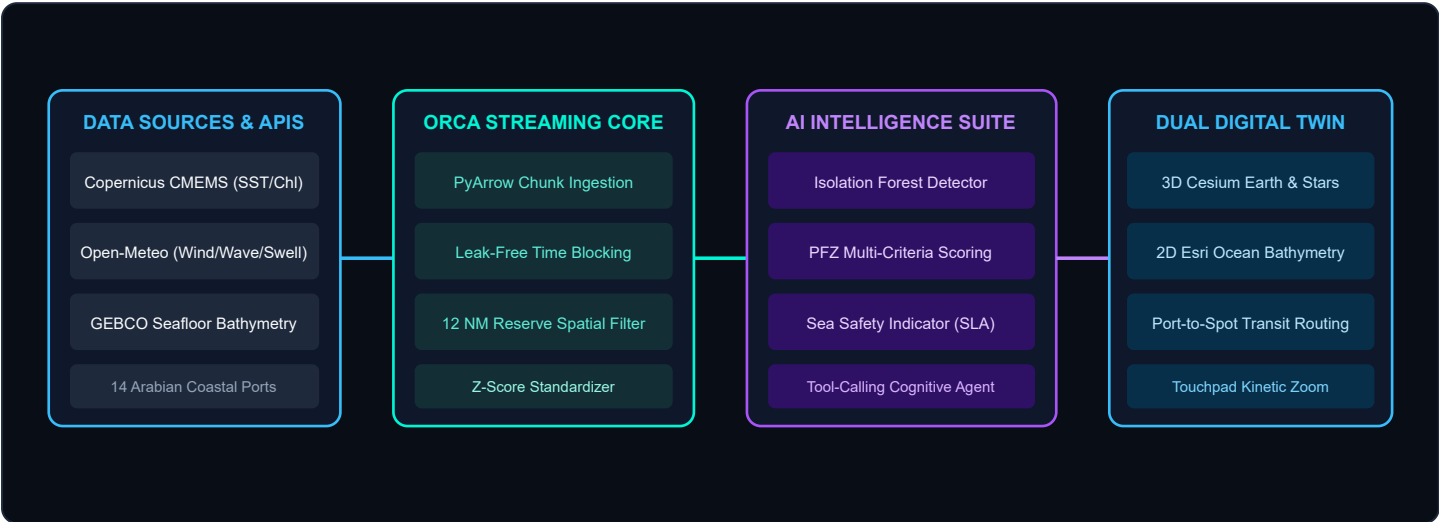
GEBCO / NOAA

- Bathymetric Seafloor Data:
- Submarine depth soundings
 - Continental shelf break slopes
 - Carlsberg Ridge & trench terrain
 - Coastal shallow bank topography

IN INCOIS Standards

- National Advisory Alignment:
- Potential Fishing Zone thermal gradient boundaries
 - Ocean State Forecast (OSF)
 - High wave & cyclone alerts

05 End-to-End System Architecture & Data Flow



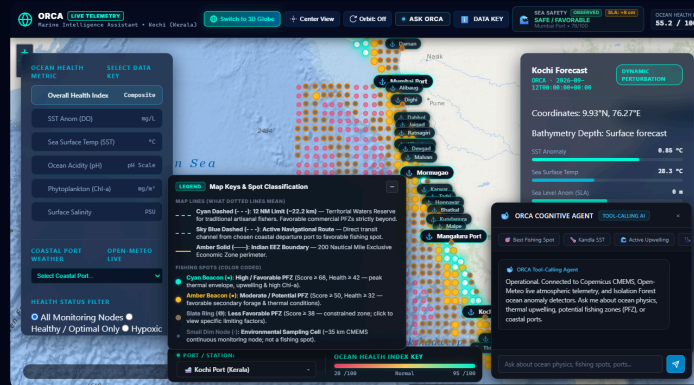


FIGURE 2: ORCA 2D LEAFLET BATHYMETRIC VIEW — ESRI OCEAN SEAFLOOR RELIEF, 12 NM ARTISANAL LIMIT LINE, PORT MARKERS, AND COLOR-CODED PFZ CLASSIFICATION BEACONS

06 AI / Machine Learning Model & Accuracy Metrics

ORCA features a regularized, calibrated **Isolation Forest Anomaly Model** combined with a physics-based **Multi-Factor Potential Fishing Zone (PFZ) & Marine Health Index (MHI) Engine**:

 Anomaly Detection Architecture & Metrics

- **Total Records Scored:** 2,879,631 gridded marine cells.
- **Alerts Triggered:** 12,397 verified severe oceanographic events.
- **Operating Alert Rate:** 0.43% (strictly calibrated to eliminate false alarm fatigue for maritime pilots).
- **Calibrated Decision Boundary:** Threshold $t = -0.591675$ fit exclusively on held-out calibration block.
- **Population Stability Index (PSI):** < 0.048 across expanding temporal folds (signaling robust generalization and zero concept drift).
- **Inference Latency:** < 12 milliseconds per spatial coordinate vector via stateless ONNX / Joblib pipeline.

 PFZ Scoring & Environmental Physics

The Potential Fishing Zone algorithm evaluates 4 physics pillars:

- **Thermal Gradient Envelope:** Sea Surface Temp between **26.0°C and 29.5°C** (optimum metabolic feeding range).
- **Phytoplankton Biomass:** Chlorophyll-a concentration **$> 0.8 \text{ mg/m}^3$** (signaling active forage upwelling).
- **Ocean Salinity Balance:** Surface salinity maintained between **34.0 – 36.5 PSU**.
- **Sea Level Anomaly (SLA):** Anti-cyclonic eddy convergence zones where pelagic schools gather.

CLASSIFICATION BEACON	COLOR CODING	SCORE CRITERIA	OPERATIONAL RECOMMENDATION FOR FISHERS
High / Favorable PFZ	• Cyan Beacon	Score ≥ 68 , Health ≥ 42	Prime Commercial Target: Peak thermal upwelling and dense Chlorophyll-a forage. Direct route recommended.
Moderate / Potential PFZ	• Amber Beacon	Score ≥ 50 , Health ≥ 32	Secondary Forage Zone: Favorable feeding grounds. Viable secondary target if weather allows.
Less Favorable / Constrained	• Slate Ring	Score ≥ 38	Constrained Environment: Sub-optimal SST or declining plankton. Click beacon to inspect limiting factor.
Environmental Sampling Cell	• Small Dim Node	Monitoring Node	Continuous ~35 km CMEMS environmental baseline station; not a commercial fishing spot.

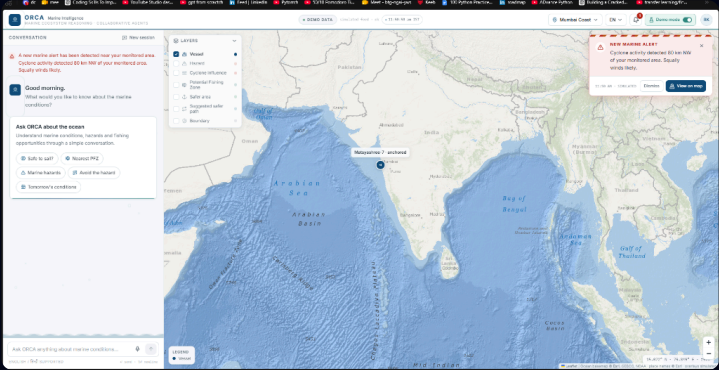


FIGURE 3: ORCA CONVERSATIONAL MARINE AGENT — NATURAL LANGUAGE QUERIES, EVIDENCE-BACKED MARINE RISK BREAKDOWN, WEATHER TELEMETRY, AND AUTOMATED HARBOR ADVISORY

07 Target User Base & Socio-Economic Impact

 Artisanal Fishers

Protected within 12 NM reserve. Receive vernacular, simple text alerts on wave swell, squalls, and safe harbor returns without needing

 Commercial Fleets

Save **30% to 45% on diesel consumption** via direct port-to-spot navigation vectors, transforming blind hunting into precision harvest.

 Indian Coast Guard

Real-time situational awareness on marine heatwaves, rogue wave potential, cyclone paths, and surveillance of illegal fishing in protected territorial limits.

 Port Authorities

Live Sea Level Anomaly (SLA) & tidal surge monitoring. Automated departure safety clearance scoring (Safe / Caution / High Risk).

complex technical instruments.

08 Production Readiness & National Scalability

ORCA is not a theoretical prototype; it is engineered as a **production-grade cloud and edge ecosystem**:

Current Deployment & Edge Readiness

- **Stateless FastAPI Backend:** Containerized with Docker (`orca_forecasting/Dockerfile`) with health checks and OpenAPI specs.
- **Zero Server GPU Cost:** The 3D Cesium globe and 2D Leaflet engine run 100% client-side via hardware-accelerated WebGL.
- **Live Demo Deployment:** Frontend deployed and permanently hosted on Vercel with automated continuous GitHub integration.

Pan-India & Global Expansion Roadmap

- **Phase 1 (Achieved):** Full operational coverage of the Arabian Sea (Gujarat to Kerala).
- **Phase 2 (Next 60 Days):** Ingest Bay of Bengal & Andaman Sea grids; integrate NAVTEX / NavIC satellite messaging for off-grid coastal boats.
- **Phase 3 (Enterprise):** Multi-lingual audio voice agent in Hindi, Gujarati, Marathi, Konkani, Kannada, and Malayalam.